

FRAD Foundation

Recruitment Platform Reference Guide

A complete description of every component of the platform: the people it serves, the process it enforces, the data it holds, and the controls that make each decision defensible.

Specification

docs/End_to_End.md

Audience

HR, hiring departments, budget holders, auditors, administrators

1. What this platform is

The FRAD Recruitment Platform manages one NGO's hiring from the moment a department realises it needs someone, to the moment the successful candidate is handed over to the organisation's existing ERP as a member of staff. It does nothing after that point.

That boundary is deliberate and it is enforced in the code, not just described in policy. The platform holds no payroll, no leave, no attendance, no performance records, no staff self-service. Once a person is transferred, their recruitment file becomes read-only and every subsequent employment matter belongs to the ERP.

The problem it solves

Recruitment in an NGO fails in predictable ways. A vacancy is advertised before anyone confirmed the money exists. Eligibility is judged inconsistently because three different officers read three hundred CVs on three different days. A criterion quietly changes halfway through. Nobody can reconstruct, a year later, why one candidate progressed and another did not.

The platform addresses each of those directly:

- **Nothing is advertised until it is funded.** A vacancy cannot be published without an approved staffing request and a Budget Holder's confirmation behind it.
- **Basic eligibility is decided by rule, not by reading.** Approved rules run automatically against every application, and reviewers only see the ones the rules could not settle.
- **The bar cannot move silently.** Longlisting rules lock at publication; changing one afterwards is an approvable proposal with a before-and-after record and, once applications exist, a fairness review.
- **Every decision has a signatory.** Approvals that confer authority are signed, and the signature is taken over a hash of exactly what was approved — so a later edit is detectable.

The shape of the process

```

Hiring department raises a staffing request
  v
Budget Holder confirms funding, budget line and ceiling
  v
HR prepares the vacancy and its longlisting rules
  v
HR Manager approves; publication locks the rules
  v
Applicants apply through the careers portal
  v
Rules run automatically; applicants are grouped
  v
HR works only the exception queue
  v
HR Manager confirms the longlist (signed)
  v
Reviewers shortlist competitively against a weighted matrix
  v
Assessments and panel interviews; scores locked on submission
  v
Scores consolidated; preferred and reserve recommended
  v
References and due-diligence checks completed
  v
Budget Holder re-confirms if the offer exceeds the ceiling
  v
HR Manager approves the offer (signed)
  v
Candidate accepts (signed)
  v
Pre-employment requirements completed and cleared
  v
HR Manager approves ERP transfer (signed)
  v
Handover pack issued; recruitment file closed and archived

```

Figure 1. The end-to-end flow. Every arrow is a state transition the platform will refuse to make out of order.

Design principles

These are the commitments the rest of this document elaborates on.

Principle	How it shows up in the platform
Built for one organisation	No multi-tenancy, no cross-organisation sharing; configuration is organisational, not per-client
Ends at ERP transfer	No staff-management modules; the record goes read-only after handover
HR Manager owns the process	Configuration, approvals and exceptions sit with HR, not with a technical administrator
Budget Holders own the money	Finance receives information but is never the business approver for recruitment funding
Longlisting is automatic and transparent	Rules are vacancy-specific, versioned, locked at publication, and every outcome names the rule that decided it
Humans handle exceptions and competition	Reviewers spend their time on unclear cases and on shortlisting, not on mechanical filtering

Principle	How it shows up in the platform
Everything important is traceable	A hash-chained audit log, entity versioning, and signatures over approved content
Nobody approves their own work	Separation of duties is enforced at every approval point
Applicant data is protected	Role-based access, restricted findings, retention rules, and anonymised early review

How to read this guide

Chapters 2 and 3 describe who uses the platform and what each of them may do. Chapters 4 to 12 walk the recruitment process in order, one stage per chapter. Chapters 10, 11, 13 and 14 cover the cross-cutting machinery: controls, data security, integrations, reporting and operations. Chapter 15 is a component index — every screen, endpoint and module, listed.

If you are implementing or administering the platform, read chapters 2, 3, 11 and 14 first. If you are an auditor, chapters 3, 10, 11 and 13 carry most of what you need.

2. The people who use it

Eleven roles exist. Each one is a set of permissions, not a job title — a person may hold more than one, and the platform cares only about what the combination allows.

2.1 Applicants

Anyone may browse published vacancies without an account. Applying requires registration and a verified email address.

An applicant can: create an account and verify it; build one reusable profile covering education, employment, licences, certifications, skills and languages; upload a CV and supporting documents; search and filter vacancies; save a draft application and return to it; submit; withdraw; track status; receive and respond to assessment and interview invitations; nominate referees; accept or decline an offer; complete pre-employment requirements; raise a complaint or appeal; request correction or deletion of their data.

An applicant can never see: internal recruitment notes, screening or interview scores, panel comments, other candidates, salary deliberations, reference responses, background-check findings, or safeguarding records. This is enforced by a separate candidate-facing status vocabulary — several distinct internal stages collapse to the same words an applicant sees, so the granularity of internal deliberation never leaks.

Internal stage	What the applicant sees
UNDER_REVIEW, EXCEPTION_REVIEW, LONGLISTED	Under review
REFERENCE_CHECK, BACKGROUND_CHECK, RECOMMENDED, RESERVE	Under review
NOT_LONGLISTED, NOT_SHORTLISTED, INELIGIBLE, NOT_SELECTED	Unsuccessful
PREBOARDING, PRE_EMPLOYMENT_CLEARANCE	Preboarding in progress
RESUMED, READY_FOR_ERP_TRANSFER, TRANSFERRED_TO_ERP	Recruitment completed

Table 1. An applicant told "under review" cannot tell whether their file was flagged as an exception. That is intentional.

2.2 Internal candidates

An internal candidate is a member of staff applying for a vacancy. They apply with the ordinary applicant role, because a recruitment role describes what someone does *in the recruitment system*, not whether they work here.

They are identified by their **verified** [@fradfoundation.org](mailto:verified@fradfoundation.org) email address. Two rules make this safe to rely on:

- The address must be verified. An unverified one proves nothing, and internal vacancies are frequently promotions or restructures that must not be visible outside the organisation.
- The domain must match exactly. not-fradfoundation.org, fradfoundation.org.attacker.com and unlisted subdomains are all rejected.

The domain list is configurable through `INTERNAL_EMAIL_DOMAINS` so a rename or a second domain does not require a code change.

2.3 Hiring department representative

The programme manager, department head or technical lead who needs the post filled.

They can: raise a staffing request with its justification; attach or confirm the job description; recommend technical eligibility and longlisting criteria; review applicants HR assigns to them; take part in shortlisting; sit on panels and submit independent scores; recommend candidates; add technical comments.

They cannot: publish a vacancy, edit an applicant's record, see confidential reference or safeguarding findings, change HR-approved longlisting rules, issue an offer, see another panel member's score before submitting their own, or approve money unless they separately hold the Budget Holder role.

2.4 Budget Holder

The person with authority over the relevant project, grant, department or budget line. **Not Finance.** Finance may receive reports, but the business decision that a post is funded belongs to the person who owns the budget.

They can: confirm that a position is funded; confirm the budget line, the funding period and whether it is grant or unrestricted; set the salary or consultancy ceiling and the maximum recruitment cost; record whether donor approval is required and its reference; re-confirm an offer that exceeds the ceiling they set; reject or return an unfunded request.

They cannot: see candidate records, scores, interview outcomes or offers beyond the financial terms. Their workspace contains no candidate data at all.

A Budget Holder cannot confirm funding for a request they raised themselves.

2.5 Recruitment officer

The day-to-day HR operator, and the busiest role in the platform.

They can: create and prepare vacancies; define longlisting rules and shortlisting matrices; run automatic longlisting; work the exception-review queue; assign reviewers; configure and schedule assessments and interviews; manage references; request and record background checks; prepare selection documentation and conditional offers; manage preboarding; record ERP handovers; handle complaints.

They cannot: approve a vacancy, confirm a longlist, reverse an automatic eligibility outcome, approve an offer, waive a required check, or approve an ERP transfer. Each of those is reserved for the HR Manager.

2.6 HR Manager

The accountable owner of the recruitment process, and the principal business approver.

They can: everything a recruitment officer can, plus approve vacancies; approve or reject staffing requests; approve changes to locked longlisting rules; confirm longlists; reverse automatic eligibility outcomes; approve selection recommendations; approve offers within delegated authority; waive required checks with a reason; approve ERP transfers; configure workflows, statuses, templates, scorecards and service standards; manage operational access for HR users; freeze or reopen stages.

The one thing they cannot do is approve their own work. A vacancy an HR Manager created is routed to a different approver.

2.7 Interview panel member

They can: see only the vacancies and candidates assigned to them; read the approved candidate pack; declare a conflict of interest; view the approved question set; enter scores and evidence-based comments independently; submit a recommendation; confirm their completed scorecard.

They cannot: see another panel member's score before submitting their own, change anyone else's score, reach unrelated vacancies, see salary details unless separately authorised, see background or safeguarding findings, or alter the final ranking.

2.8 Executive approver

The Executive Director, Country Director or a delegated senior officer. They are involved only where policy requires it: senior-management recruitment, vacancies created by the HR Manager that need second-level approval, recruitment outside the approved structure, exceptional salary proposals, policy waivers, direct appointments, cancellations at an advanced stage, and emergency recruitment.

The platform decides when to escalate rather than leaving it to judgement. A staffing request routes to an executive automatically when the grade is senior, the urgency is emergency, or a new establishment of three or more posts is proposed — and the reason for the escalation is displayed alongside it.

2.9 Referee

An external person completing a reference. They reach the platform through a single-use, expiring link and never hold an account. They see the questions and nothing else — no internal records, no other referees' responses, no candidate file.

2.10 Course administrator

Manages preboarding course content, quiz authoring, enrolment visibility and attempt resets. Sees nothing else.

2.11 Auditor

Read-only access to recruitment records, the audit trail and authorised report exports. Cannot mutate configuration or make any recruitment decision.

2.12 Technical system administrator

Manages the technology, not the recruitment process.

They can: create and disable accounts; assign roles on approved instruction; configure authentication; manage backups, integrations and email delivery; monitor availability; manage security settings and software updates; restore from backup where authorised.

They cannot: approve vacancies or candidates, alter scores, change selection decisions, issue offers, approve funding, or change any recruitment outcome. This is enforced structurally: a system-administrator account is restricted to three technical permissions and *cannot inherit recruitment authority even if an operational role is mistakenly also assigned to it*. Someone who does both jobs needs two accounts, so every decision has an unambiguous actor.

All technical-administrator access to recruitment data is logged.

2.13 Separation of duties, summarised

Decision	Who may make it	Who is excluded
Raise a staffing request	Hiring department, HR	—
Confirm funding	Budget Holder	Anyone who raised the request; Finance
Approve a vacancy	HR Manager	The person who created it
Change a locked rule	HR Manager	The person who proposed it
Confirm a longlist	HR Manager	—
Reverse an automatic outcome	HR Manager	Recruitment officers
Score an interview	Assigned panel members	Anyone unassigned
Approve an offer	HR Manager, executive	The person who prepared it
Approve an ERP transfer	HR Manager	—
Waive a required check	HR Manager	Recruitment officers

3. Staffing requests and funding

Every vacancy starts here. There is no way to create a publishable vacancy without an approved staffing request behind it.

3.1 What a request captures

A request records twenty-five fields across five groups. The grouping matters because different people own different parts.

Group	Fields
Position and structure	Position title, department, project or grant, duty station, number of positions, replacement or new, previous holder, reason for recruitment, proposed reporting line
Contract and timing	Contract type, proposed duration, expected start date, job grade, urgency
Funding (proposed)	Budget line, funding source, funding end date, proposed salary ceiling, donor restrictions
Requirements	Job description attachment, required qualifications, required experience, required languages, safeguarding sensitivity, proposed assessment method, proposed panel
Requester	Hiring manager name, email, phone

Two validation rules run at entry:

- A replacement must name the previous holder, otherwise the establishment record cannot be reconciled.
- A grant-funded position must carry a funding end date, and that date must fall after the expected start — a post funded only up to the day it starts is not funded.

3.2 The lifecycle

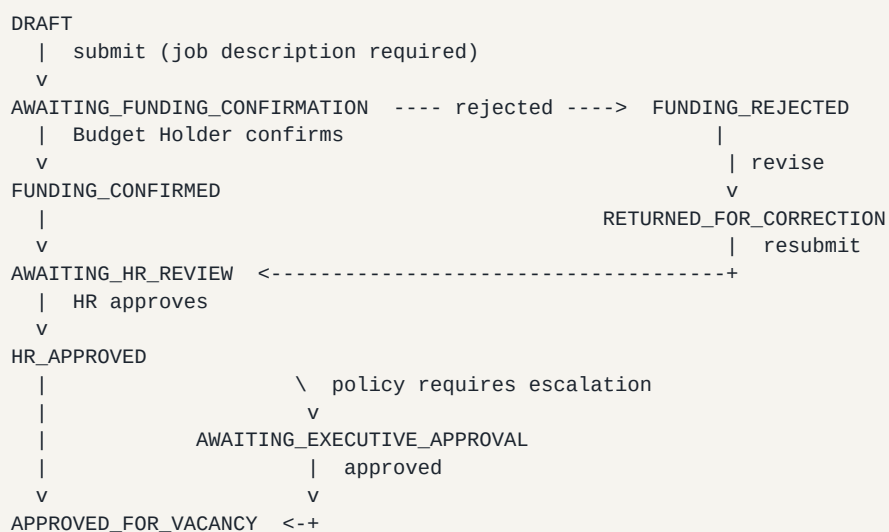


Figure 2. Twelve statuses. The transition table refuses anything not drawn here — a request cannot reach approval without passing through funding.

The ordering is the point. Funding is confirmed *before* HR spends time reviewing the substance, because a request nobody will pay for should not consume HR's attention.

3.3 Submission

Submitting requires an attached job description. The platform blocks submission without one, because the job description is what HR, the panel and the eventual scorecard all recruit against — reconstructing it later from the request narrative produces a different job.

On submission the requester signs. The signature covers the position title, number of positions, budget line and grade, so a later quiet edit to any of those is detectable.

3.4 Automatic escalation

The platform decides whether an executive must be involved, rather than leaving it to whoever is handling the request.

Trigger	Reason shown
Job grade begins D, E, SM or EX	Senior management grade
Urgency is Emergency	Emergency recruitment
Not a replacement, and three or more positions	New establishment of three or more positions

A routine single replacement at a junior grade does not escalate. When escalation does apply, the approval is routed to an approver who is independent of both the requester and the HR reviewer, and the reason is displayed alongside the request so nobody has to guess why it stalled.

3.5 The funding decision

This is the Budget Holder's screen, and it is deliberately narrow — it contains no candidate information whatsoever.

A confirmation records the budget line, funding source, funding start and end, the salary or consultancy ceiling with its currency, the maximum recruitment cost, whether the post is grant-funded, whether donor approval is required and its reference, and a free-text comment.

Confirming requires the budget line, the ceiling and the funding end date. A confirmation without them is not a confirmation — it is an opinion. Rejecting or returning requires a reason.

Only one confirmation is ever current. Recording a new one supersedes the previous, which is retained for audit. This means the question "what is the approved ceiling for this post?" always has exactly one answer, and the history of how it changed is still available.

The decision is signed by the Budget Holder, and the signature is written in the same database transaction as the confirmation itself. If the signature cannot be written, no funding decision is recorded at all — there is no

state in which an approved envelope exists without a signatory.

On confirmation the request moves straight to HR review, and the confirmed budget line and funding end date are copied onto the request so the vacancy inherits the real figures rather than the proposed ones.

3.6 Offer variations

When an offer is later prepared above the confirmed ceiling, it is flagged and returned to the Budget Holder for re-confirmation before the HR Manager may approve it. These appear on the Budget Holder's dashboard as a distinct queue.

3.7 The Budget Holder dashboard

Four sections, and nothing else:

- Staffing requests awaiting a funding decision, with emergency requests highlighted
- Offers that exceed a confirmed ceiling and need re-confirmation
- The Budget Holder's own decision history, including superseded ones
- Funding ending within three months, so a post is not filled against money about to expire

3.8 Creating the vacancy

A vacancy may be linked to a staffing request at any point while it is a draft, but the link is validated: only a request in [APPROVED_FOR_VACANCY](#) may be used. Publication then checks both the link and the funding behind it.

4. Vacancies

4.1 What a vacancy record holds

Group	Fields
Identity	Reference number (unique, auto-assigned), title, department, project, category, duty station
Establishment	Number of openings, reporting line, grade, contract type, contract duration
Content	Summary, responsibilities, essential criteria, desirable criteria, desired experience, language requirements, technical skills, behavioural competencies, safeguarding responsibilities, travel requirement
Timing	Opening date, closing date, time zone
Pay	Salary disclosure setting, range minimum and maximum, currency
Governance	Safeguarding classification, recruitment contact name and email, originating staffing request, owner
Assessment	Application questions, required documents, screening scorecard, interview scorecard, preboarding package
Special routes	Audience, emergency flag and justification, anonymised review policy
Locking	Timestamp at which longlisting rules locked

Salary disclosure has three settings — hidden, range, or exact. Choosing *range* requires both bounds, and the maximum may not fall below the minimum.

4.2 Audience: public and internal vacancies

Every vacancy is **PUBLIC**, **INTERNAL** or **BOTH**.

An internal vacancy is visible only to a verified member of staff. This is enforced in four places, not one:

- The careers page filters by audience against the viewer's identity
- The public vacancy list endpoint does the same
- The public vacancy detail endpoint does the same, so an internal reference number cannot be resolved by guessing it
- The application endpoint blocks submission *and* draft saving, so nobody can start an application against a role they are not eligible for and submit it later

Where a member of staff is selected for an internal vacancy, the result still produces the standard handover pack, flagged as a transfer or promotion action rather than a new hire.

4.3 Application questions

Ten question types are available: text, long text, number, date, yes/no, single select, multi select, file upload, declaration, and conditional questions that appear only when an earlier answer requires them. Conditions are enforced on the server, so a hidden question cannot be answered by manipulating the form.

Selection questions require at least two options. Every question requires a label. A vacancy may hold at most one hundred questions.

4.4 Required documents

Each required document specifies its type, whether it is mandatory, the accepted file extensions, a maximum size, and whether an expiry date must be supplied. Document types are configured centrally, so a vacancy cannot request a type the organisation does not recognise, and each type may be requested only once.

4.5 Approval and publication

A vacancy moves **DRAFT** to **PENDING_APPROVAL** to published. Submitting for approval routes it to an HR Manager who is independent of the person who created it.

Publication is gated. The platform refuses to publish until every one of these is true:

Check	Why
A staffing request is linked	A vacancy with no origin has no justification
That request is approved for vacancy preparation	An unapproved request is still a proposal
Budget Holder funding is confirmed	Nothing is advertised that is not funded
At least one mandatory longlisting rule exists	Otherwise automatic longlisting can decide nothing
A screening scorecard is chosen	Shortlisting criteria must exist before applications arrive
An interview scorecard is chosen	The assessment structure must be settled in advance
At least one application question exists	
Safeguarding classification is set	It determines which due-diligence checks apply
A recruitment contact email is recorded	Applicants need a real person to ask
Mandatory content is complete	Reference, title, summary, responsibilities, essential criteria, positions, valid dates
Screening scorecard weights total 100	A matrix that does not sum is not a matrix

When the vacancy is emergency-classified, three further checks apply — see chapter 12.

The response lists every unmet condition at once rather than reporting them one at a time.

Publication locks the longlisting rules. From that moment a rule change is no longer an edit; it is a proposal requiring HR Manager approval, carrying a before-and-after record, and triggering a fairness review if applications have already been received. See chapter 6.

4.6 Vacancy statuses

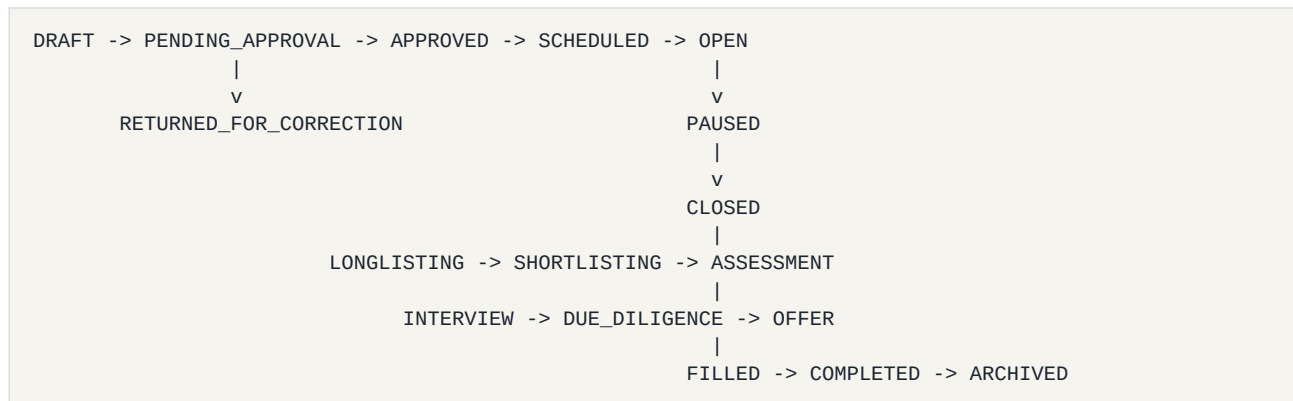


Figure 3. The vacancy tracks which recruitment stage it has reached, so a portfolio view shows where each exercise actually is.

A vacancy may be paused with a reason, extended with a later closing date and a reason, cancelled by an HR Manager with a reason, or duplicated to seed a new draft — the duplicate copies content, questions and document requirements but never the approval, the applications or the lock.

4.7 The careers portal

Public pages list open vacancies with search and filtering by department, category and duty station. Search terms are capped at one hundred characters, because the parameter is untrusted and drives multiple pattern comparisons.

Each vacancy page shows the summary, responsibilities, criteria, duty station, closing date and recruitment contact, and — where disclosure allows — the salary range. It also carries the privacy notice and the fraud warning: FRAD never charges a fee at any stage of recruitment.

The portal is mobile-responsive, works on low bandwidth, saves incomplete applications, and states deadlines in the vacancy's own time zone.

5. Candidates and applying

5.1 The reusable profile

A candidate maintains one profile and reuses it across every application. It holds personal and contact details, work authorisation and availability, and repeating records for education, employment, licences, certifications, skills and languages, plus a document library.

Each employment entry carries employer, job title, employment type, country, dates, responsibilities, reason for leaving and supervisor contact — including whether the candidate permits that supervisor to be contacted, which the reference stage respects.

Licences carry a verification status. This matters at longlisting: an unverified licence is treated as *unclear* rather than as a failure, so a genuine professional is not rejected because HR has not yet checked their certificate.

5.2 CV parsing

When a candidate uploads a CV the platform extracts what it can: name, email, phone, education entries with qualification, field of study and year, employment entries with dates and current-role detection, skills, certifications, languages, professional memberships, and an estimate of total experience that merges overlapping roles rather than adding them up.

Two rules govern the parser:

- **It only organises information.** It never scores, ranks or judges suitability.
- **The candidate reviews and corrects everything** before the application is submitted. What the parser produces is a draft.

The parser is deliberately conservative. Where a line is ambiguous it omits the field rather than guessing, because a wrong value the candidate has to notice and correct is worse than a blank one they expect to fill. A CV with no recognisable structure yields nothing and reports low confidence, rather than inventing entries.

5.3 Making an application

The application combines profile data with vacancy-specific answers. Candidates may save a draft repeatedly and return to it; drafts auto-save and recover locally if a connection drops mid-form.

Submission requires:

- A verified email address
- Every mandatory question answered, with conditional questions evaluated server-side
- Every mandatory document supplied
- The declarations accepted: conflict of interest, relationship to NGO staff, safeguarding, data-processing consent, and accuracy

One application per candidate per vacancy is enforced by a database constraint, not by a check that could race.

On submission the platform takes a **profile snapshot**. This is what the rest of the process assesses. A candidate who edits their profile afterwards does not change the basis on which their application was judged, and a longlisting run repeated months later produces the same result.

5.4 What the candidate sees afterwards

A dashboard listing their applications with a plain-language status, what it means, what they should do, and what happens next. Alongside it: tasks, messages, interview invitations, assessment invitations, document requests, offers and preboarding.

Every status carries specific guidance rather than a generic placeholder. "Under review" tells them no action is needed; "Incomplete" tells them exactly what is missing and that the deadline still applies.

5.5 Accessibility and low connectivity

- Keyboard-navigable throughout, with landmarks and accessible names
- Reduced-motion respected
- Responsive layouts down to small screens
- Print styles for offline reference
- Plain-language errors, never raw validation codes
- Auto-save, local draft recovery, retryable uploads
- HR-assisted entry for candidates who cannot apply online, recorded as assisted and audited

5.6 Complaints and appeals

An applicant may raise a complaint or request a review of the process. The platform records the vacancy, the candidate, the nature of the complaint, the date received, supporting documents, the assigned reviewer, a conflict check on that reviewer, findings, the decision, the communication sent and the closure date.

The appeal process never exposes information about other candidates.

5.7 Privacy rights

Candidates may export their data and request correction or deletion. Deletion requests are reviewed rather than executed blindly, because a recruitment record under legal hold or within a statutory retention window cannot simply be erased — the platform records the decision and its basis either way.

6. Longlisting

Longlisting decides who is *eligible*. Shortlisting, later, decides who is *best*. Keeping those separate is what lets the first be automatic and the second stay human.

6.1 Rule types

Eighteen rule types are available, covering every criterion the specification calls for.

Rule	What it evaluates
Minimum academic qualification	Highest qualification held, compared on a rank ladder
Required field of study	Field of study across all education entries
Minimum years of experience	Total employment, overlaps merged
Minimum NGO experience	Employment at recognisably humanitarian organisations
Minimum technical experience	Employment matching configured keywords in title or duties
Minimum management experience	Employment indicating supervisory responsibility
Required professional licence	Verified, unexpired licences
Required certification	Current certifications
Required language	Language at or above a configured level
Required computer skill	Skill at or above a configured proficiency
Required sector experience	Sector signals across employer, title and duties
Duty-station acceptance	Relocation willingness or stated location preferences
Work authorisation	Right to work
Availability before a date	Earliest start date
Willingness to travel	A specific application answer
Mandatory document	Documents supplied
Mandatory question	A question was answered at all
Specific answer	An answer compared with equals, in, contains, true, at-least, at-most, before or after

6.2 Rule classification

Each rule is marked with how it behaves when unmet. This is the difference between a criterion that disqualifies and one that merely informs.

Classification	Effect
Mandatory knockout	Failure makes the applicant automatically ineligible

Classification	Effect
Scored	Contributes a weighted score; never disqualifies
Preferred	Recorded for the shortlist; never affects the longlist
Informational	Captured for reporting only

6.3 Four outcomes per rule, not two

Every rule returns one of four results:

- **Met** — the requirement is satisfied
- **Not met** — it demonstrably is not
- **Unclear** — the evidence does not settle it
- **Not applicable** — the rule does not apply to this application

Unclear is the important one, and it is what makes the engine safe. Examples of where it is returned rather than a rejection:

- A qualification the ladder does not recognise, such as a foreign award — this is the "equivalent qualification" case a human must decide
- A licence that is held but not yet verified
- A certification held but expired
- A candidate who stated no location preference at all
- A skill claimed with no stated proficiency
- Any rule type the engine does not recognise

An unrecognised rule type never silently passes an applicant. It produces *unclear* and goes to a human.

6.4 How an application is placed

```

Is the application complete? -- no --> INCOMPLETE_APPLICATION
    | yes
    v
Is it a second application from the same candidate? -- yes --> DUPLICATE_APPLICATION
    | no
    v
Did any mandatory knockout return NOT MET? -- yes --> AUTOMATICALLY_INELIGIBLE
    | no                                     (deciding rule recorded)
    v
Did any mandatory knockout return UNCLEAR? -- yes --> REQUIRES_REVIEW
    | no                                     (deciding rule recorded)
    v
Are there any mandatory rules at all? -- no --> REQUIRES_REVIEW
    | yes
    v
AUTOMATICALLY_ELIGIBLE

```

Figure 4. Precedence is deliberate: a definite failure outranks an unclear result, and an empty rule set can never make anyone automatically eligible.

Preferred and informational rules never change the outcome. A scored rule contributes to the eligibility score but cannot disqualify.

6.5 The eligibility score

Where scored rules are configured, the platform computes an eligibility score as the weighted sum of those that were met, alongside the maximum available. Where none are configured the score is reported as absent rather than as zero, because those mean different things.

6.6 Running a longlist

A run may be triggered manually, on deadline close, or as a re-run. It requires at least one mandatory rule and a published vacancy — running against a draft would evaluate nothing while implying a result.

The run:

1. Supersedes any earlier unconfirmed run, so only one proposal is ever live
2. Snapshots the rule set, making the run reproducible
3. Identifies duplicate candidates with a single grouped query
4. Evaluates every application in cursor-paged batches, writing results in bulk per page
5. Records, for each application, the per-rule outcomes, the score, and the rule that decided it
6. Produces the summary

A single unparseable application is recorded as requiring review rather than abandoning the run — the safe direction.

6.7 The summary

Counter	Meaning
Total applications	Everything not still in draft
Complete / incomplete	Whether mandatory documents and answers were supplied
Automatically eligible	Cleared every mandatory rule
Automatically ineligible	Failed at least one
Requires review	At least one mandatory rule was unclear
Duplicates	A second application from the same candidate
Reason distribution	Count per rule, so the most common blocker is visible

The reason distribution is the most useful number in the platform for improving a vacancy: if one rule accounts for eighty percent of rejections, either the criterion or its wording is wrong.

6.8 The exception queue

Reviewers see only what the engine could not settle. For each, the queue shows every rule with what was expected, what was found and the outcome, with the deciding rule highlighted.

A decision requires:

- One of three outcomes: eligible, not eligible, or more information needed
- An approved **reason code**, not free text alone
- A written justification of at least ten characters
- Supporting evidence, where the reason code asserts a fact the engine could not see

Reason code	Evidence required	Approval required
Equivalent qualification	Yes	Yes
Equivalent experience	Yes	No
Missing document since received	Yes	No
Approved policy exception	Yes	Yes
Data entry corrected	No	No
System parsing error	No	No
Reasonable accommodation	No	No
Duplicate resolved	No	No

6.9 Overrides

Resolving an *unclear* case is ordinary review work. **Reversing a definite automatic outcome is not** — turning an automatically ineligible applicant into an eligible one, or the reverse, requires override authority held by the HR Manager.

The original automatic result is written once, at evaluation time, and is never modified. An override sits alongside it. Both are shown in the queue and in the exceptions report, so it is always possible to see what the rules said and what a human decided instead.

6.10 Rule locking

Before publication, rules are freely editable and every change is audited.

After publication they lock. A change becomes a proposal that:

- Requires HR Manager approval, from someone other than the person proposing it
- Records the complete before and after
- Records how many applications existed at the time
- **Triggers a mandatory fairness review** if that count is above zero, and the approval cannot complete without a written fairness note

Adding an entirely new rule after publication is refused outright — that is a different vacancy, and it should be one.

6.11 Confirming the longlist

Confirmation is what produces the longlist. Until then the outcome is a proposal and no application has moved.

The platform refuses to confirm while any exception remains undecided. On confirmation it moves each application to the stage its outcome implies, respecting any human decision that overrode the automatic one, and never dragging an application backwards out of a later stage it has since reached.

The confirmation is signed by the HR Manager, inside the same transaction as the stage changes. If the signature cannot be written, none of the moves happen either. There is no state in which a longlist exists without a signed approval behind it.

7. Shortlisting, assessment and interviews

7.1 Shortlisting

Longlisting established eligibility. Shortlisting compares eligible applicants against one another using a weighted matrix defined before applications arrived.

Each criterion carries a definition, the evidence expected, a weight, a maximum score, a minimum acceptable score, and space for the reviewer's score and comment. Criterion weights must total 100 — the platform blocks publication otherwise, because a matrix that does not sum produces scores that cannot be compared.

The reviewer process supports one or more reviewers per candidate, independent scoring, candidate assignment, reviewer deadlines, conflict-of-interest declarations, score variance alerts where reviewers diverge sharply, a consolidated score, and final approval.

Controls: reviewers cannot change longlisting rules; they may score only eligible candidates; scores lock on submission; a change afterwards requires a reason and preserves every prior version.

7.2 Anonymised review

A vacancy may be configured to hide identifying information during early review. HR chooses which fields, per vacancy:

Name, photograph, gender, age and date of birth, address, nationality, religion, marital status, contact details, institution names.

Where the name is hidden, reviewers see a **stable alias** — "Candidate 7C3A" — derived from the application reference, so it stays consistent across screens and between sessions and colleagues can discuss a candidate without knowing who they are.

Three properties make this trustworthy:

- **Redaction happens on the server**, before the data is serialised. A value the browser never receives cannot be revealed by inspecting the page.
- **Free text is scrubbed too**. A motivation statement that names the candidate, or contains their email or phone number, is redacted — otherwise the whole exercise fails on the first personal statement.
- **It fails closed**. If the configuration is unreadable, the platform hides the full default set rather than revealing everything.

Anonymisation applies to screening and shortlisting only. Once a candidate reaches interview, references or offer, identity is unavoidable and hiding it would obstruct the process rather than reduce bias.

7.3 Assessments

Ten assessment types are supported: written tests, technical tests, case studies, presentations, practical tests, computer tests, language tests, driving tests, data exercises and clinical scenarios.

Each assessment carries instructions, a time limit, candidate-specific links, a question bank, file-upload tasks, pass marks, reviewer assignment, late-submission rules and accommodation settings. Objective questions can score automatically; everything else is marked with recorded evidence.

Timed assessments auto-save as the candidate works and auto-submit at expiry, so a lost connection near the deadline does not lose the attempt. Attempt resets require a reason and are audited.

Results lock on submission and require approval before they influence ranking.

7.4 Question and assessment banks

HR maintains a controlled bank of eligibility questions, longlisting questions, technical test questions, competency questions, safeguarding questions, interview questions, scoring guides and model answers.

The bank supports categories, difficulty levels, job families, version control, restricted access, random selection, reuse across vacancies, and review and expiry dates — so a question set does not silently age into irrelevance.

7.5 Interviews

Six formats: panel, technical, competency, final, remote and physical.

The flow covers panel nomination and approval, conflict declarations, candidate scheduling, calendar invitations, reminders, the interview guide, standard and technical and competency questions, a mandatory safeguarding question, the scoring rubric, independent scoring, the consolidated result and the panel recommendation.

Panel members cannot see one another's scores until all required scores are submitted. This is the single most important control in the interview stage, and it is enforced server-side rather than by hiding a column in the interface.

A submitted scorecard locks. Reopening one requires the reopen permission and a recorded reason, and the prior version is retained.

7.6 Scheduling and calendar integration

The platform integrates with Microsoft Outlook and Teams, Google Calendar and Meet, and Zoom, through OAuth with PKCE. Each provider reports itself unavailable until an administrator configures its credentials, rather than failing at the point a user tries to connect.

Once connected, the platform can:

- Read free/busy time to establish panel availability
- Propose interview slots where the panel is genuinely free
- Create the calendar event and provision the meeting link
- Send invitations, reminders and reschedule notices
- Record attendance confirmation

Slot proposal respects the **vacancy's own time zone**, not the server's or the coordinator's. Working hours and weekends are evaluated in that zone, so a Friday-evening slot in one country is not proposed as a Saturday-morning slot in another. Daylight saving is handled correctly because offsets are resolved per date rather than assumed.

Panel members who declared explicit availability windows are only offered within them; those who declared nothing are treated as open. Slots are ranked by how much of the panel can attend, then by earliest.

Tokens are sealed with AES-256-GCM before storage and refreshed lazily on read. The PKCE verifier is sealed the same way.

7.7 Ranking and recommendation

Approved stage scores are combined by configured weights — for example shortlisting 20 percent, written test 30 percent, interview 50 percent.

The ranking shows each candidate's stage scores, weighted total, rank, pass or fail status, the panel recommendation, and any disqualification reason. It identifies a preferred candidate and a reserve.

Where the highest-ranked candidate is not the one recommended, the platform requires a written justification. That justification appears in the selection report and is one of the first things an auditor looks for.

Selection decisions are approved through maker-checker: the person who created the recommendation cannot approve it.

8. References and due diligence

8.1 References

Referees are nominated by the candidate and contacted through a secure, single-use, expiring link. A referee never holds an account and never sees anything except the questions.

The platform supports structured reference requests, reminders, verification of the response, records of referees who could not be contacted, manager-authorised waivers, and HR-recorded manual references where a referee replied by phone or letter. A manual entry requires evidence and must resolve to one of the canonical outcomes — satisfactory, concerns raised, or unsatisfactory — so it cannot be recorded as an ambiguous note.

Reference responses may carry a confidential comment visible only to authorised HR. Panel members never see reference content.

8.2 Background and due-diligence checks

Nine check types are supported:

Identity verification, qualification verification, employment verification, professional licence verification, criminal-record check, sanctions screening, safeguarding check, work-authorisation check, and driving-licence verification.

Which checks apply

The required set is derived from the vacancy rather than chosen ad hoc:

Condition	Checks added
Every vacancy	Identity, qualification, employment, work authorisation
Safeguarding classification elevated or high	Safeguarding, criminal record
Safeguarding classification high	Sanctions screening
Title indicates a driving role	Driving licence
Title indicates a clinical role	Professional licence

Lawful basis

Criminal-record, safeguarding and sanctions checks are only lawful in particular circumstances. The platform requires a written lawful basis to be recorded **before** such a check may be requested — not afterwards as documentation.

Data minimisation

Each check type declares the minimum field set it needs, and only those fields are transmitted to a provider.

Check	Fields shared
Identity	Full name, date of birth, identity document number
Qualification	Full name, institution, qualification, completion year
Employment	Full name, employer, job title, start and end dates
Professional licence	Full name, professional body, licence number
Criminal record	Full name, date of birth, identity document number
Sanctions screening	Full name, date of birth, nationality
Safeguarding	Full name, date of birth
Work authorisation	Full name, nationality, permit number
Driving licence	Full name, licence number, licence class

The audit log records **which fields were shared**, never their values.

Restricted findings

Findings for criminal-record, safeguarding and sanctions checks are restricted. Everyone managing the process can see that such a check exists and what state it is in — otherwise the process could not be managed — but the finding text, restricted notes and evidence file are removed **on the server** for anyone without the restricted-read permission.

Writing a finding on a restricted check requires the same authority as reading one; read and write are not separable here.

Recording *concerns raised* requires a written summary of the concern. A status alone is not a finding.

Waivers

A required check may be waived, but only by an HR Manager, and only with a reason of at least fifteen characters. Waivers appear in their own report.

Check states

State	Meaning
Not requested	Identified as required, not yet initiated
Requested	Sent to the provider or internal team
In progress	Provider acknowledged
Response received	Returned, not yet reviewed
Cleared	No adverse finding
Concerns raised	Adverse or ambiguous; requires a written summary
Failed	Adverse and disqualifying
Waived	Explicitly waived by an HR Manager with a reason
Not applicable	Determined not to apply

Anything other than cleared, waived or not-applicable blocks the offer.

8.3 Document verification

Documents supplied by candidates carry an authenticity status, verification notes, expiry dates, the verification source, who verified them and when, a rejection reason where applicable, a replacement request, restricted access, and full version history — so a re-uploaded certificate does not erase the record of the one it replaced.

9. Offers, preboarding and ERP handover

9.1 Offers

An offer records the position, duty station, contract type and duration, proposed grade and step, salary amount with currency and period or a consultancy fee, budget line, funding source and end date, donor restriction, start and end dates, probation, reporting line, conditions, and the acceptance deadline.

HR prepares it. The **Budget Holder** confirms the financial envelope. The **HR Manager** approves within delegated authority. Where the proposed amount exceeds the confirmed ceiling the offer is flagged and returned to the Budget Holder for re-confirmation before HR Manager approval is possible.

Finance does not replace the Budget Holder in this workflow at any point.

Offers are versioned. A revised offer supersedes its predecessor, and both are retained.

The candidate's side

The candidate can view and download the offer as a PDF, request clarification, propose a different start date, accept by typing their full legal name as an electronic signature and confirming they have read the terms, upload a countersigned PDF, or decline. Unanswered offers expire at the deadline.

Acceptance captures the typed name, the method, the IP address and the user agent, and is signed — a critical signature, so an acceptance that cannot be signed is not recorded at all.

9.2 Preboarding

Once an offer is accepted the candidate receives a preboarding package: a versioned bundle of forms, document requirements, policies, courses, tasks and meetings, materialised as a snapshot so a later change to the template does not alter what an in-flight candidate was asked to do.

Component	What it covers
Forms	Dynamic forms with auto-save, submission, HR review, approval, return or waiver
Documents	Versioned uploads with verification, expiry and rejection reasons
Policies	Acknowledgement by tick, typed name, drawn signature or signed PDF
Courses	Versioned content, quizzes, attempts, reasoned resets, grading, downloadable certificates
Tasks	Assigned pre-start actions with due dates
Meetings	Scheduled orientation with attendance confirmation
Information items	Reporting instructions and travel details

Pre-employment requirements cover accepted offer, signed contract, identification, academic certificates, professional licence, references, background checks, safeguarding clearance, work permit, medical fitness where lawful and job-related, bank details, tax details, pension details, emergency contact, next of kin, code of conduct, PSEA declaration, conflict-of-interest declaration and confidentiality agreement.

Each item carries one of eight states: not requested, requested, submitted, under review, verified, rejected, waived with approval, or not applicable.

Readiness checks run against the mandatory set. A waiver requires a reason. Clearance is the gate that precedes readiness to start, and it is an HR Manager act.

9.3 Resumption

HR records the confirmed start date, the reporting location, and afterwards the actual outcome — resumed, did not resume, postponed or withdrawn — with the actual start date.

9.4 ERP handover

Transfer is **manual by design**. The platform does not write to the ERP. What it produces is a single authoritative document containing exactly the data the ERP needs, so the person keying it in has one accurate source and the recruitment file records precisely what was handed over.

Readiness

Approval is refused until every one of these holds:

- An accepted offer exists
- Preboarding was assigned
- Pre-employment clearance was issued
- Every mandatory background check is cleared, waived or not applicable
- A confirmed start date exists

Where actual resumption has not been recorded the platform warns rather than blocks, and the pack shows the planned start date instead.

Duplicate employee check

Before approval, the platform looks for an existing employee who may be the same person. It compares only against people who **actually hold an ERP number** — a transfer still in flight is not a duplicate, and matching against one would flag every candidate against themselves.

Identity, email address and phone number each block on their own. A shared full name is reported as context but does **not** force an override, because common surnames would otherwise produce constant false positives and the written override would become a formality people click past.

Where a blocking match is found, approval requires both an explicit acknowledgement and a written explanation.

Approval

The HR Manager approves. The approval record, the stage change and the signature are written in one transaction — an unsigned ERP transfer approval cannot exist.

The handover pack

A branded PDF containing:

Section	Contents
Transfer authorisation	ERP number or a statement that one is to be assigned, recruitment and vacancy references, approver, approval date, generation timestamp, duplicate-check outcome
Personal details	Full name, preferred name, email, phones, address, nationality
Position and contract	Department, position, grade, step, supervisor, duty station, contract type, start and end dates, budget or project code
Remuneration	Salary with currency, period, budget line, funding source
Statutory and payroll	Bank, tax, pension, emergency contact, next of kin
Clearance evidence	Offer acceptance date and signature, clearance date, every check with its status and outcome

Statutory data is drawn from the preboarding form **explicitly designated** to supply it, through a configured purpose code — never by matching a form title, because renaming a form would then silently empty the pack.

The pack distinguishes three states that would otherwise all read as blank:

- **Not collected** — no form is configured to collect this at all
- **Outstanding** — a form exists but the candidate has not submitted it
- **Error** — the stored response could not be read

These require different responses from HR, and the gaps are surfaced on the approval screen before the pack is issued rather than discovered by payroll afterwards.

Every download is audited, and the transferred dataset is snapshotted so the handover can be reconstructed later.

Recording the transfer

Once the ERP issues a personnel number, HR records it. The field holds only a real number — it is null until then, and workflow state lives in a separate status. Recording requires prior approval and is idempotent, so a retried request cannot create a second transfer.

After transfer

The candidate's status becomes transferred to ERP. The recruitment record becomes read-only except for authorised closure actions. All ongoing staff management happens in the ERP. The recruitment file is retained according to the retention policy.

10. Controls, audit and signatures

10.1 The audit log

Every consequential action writes an entry recording the actor, the action, the resource, the previous and new values, a reason where one was supplied, the IP address, the user agent and the timestamp.

The log is **hash-chained**. Each entry includes the hash of the one before it, and the chain head is stored separately. Verification walks the whole chain and reports one of several specific failures rather than a generic invalid:

Result	Meaning
Hash mismatch	An entry's content no longer matches its own hash — it was edited
Missing previous entry	An entry was deleted from the middle
Chain branch or missing head	More than one entry claims to be the head
Chain cycle	The chain loops
Disconnected chain	Entries exist outside the chain
Head pointer mismatch	The stored head does not match the computed one

Writing is retried on contention, and a persistent failure raises a critical operational event rather than passing silently.

10.2 Electronic signatures

Sixteen document classes can be signed: staffing requests, vacancy approvals, conflict declarations, shortlist approvals, longlist approvals, interview scorecards, selection recommendations, reference forms, funding confirmations, offer approvals, conditional offers, final offers, candidate acceptances, pre-employment declarations, ERP transfer approvals, and recruitment closures.

Each signature records the signatory, their role, the method, how they were authenticated, the document version, a hash of exactly what was approved, the status, the IP address, the user agent and the timestamp.

Why the hash matters

The signature is taken over a canonical hash of the approved content. Canonical means key order does not affect the result, so the same logical payload always hashes identically — but any actual change does not:

- Changing a value changes the hash
- Adding a field changes the hash
- A null and an absent field hash differently
- Array order is significant
- The number 1 and the string "1" hash differently

This is what makes a silent change *detectable* rather than merely discouraged. Re-verifying a signature against the current record answers "is what was signed still what is there?"

Critical signatures are blocking

Thirteen of the sixteen classes are **critical** — those that confer legal, financial or safeguarding authority. For these the signature *is* the record of authority, so:

- The signature is written **inside the same database transaction** as the state change it attests
- If it cannot be written, the entire transaction rolls back
- The caller receives a specific error saying nothing was saved and the action should be retried

There is no state in which a funding confirmation, longlist confirmation, offer approval, candidate acceptance or ERP transfer approval exists without a signature behind it.

The three non-critical classes — conflict declarations, reference forms and pre-employment declarations — degrade gracefully instead, logging a high-severity operational event, because an informational acknowledgement should not take down the action it accompanies.

Amendments

A signature is never deleted. Superseding one marks the original as amended, links it forward to its replacement, and records the reason.

10.3 What the platform prevents

Prevented	Mechanism
Self-approval	Independent approver selection excludes the requester, and route handlers re-check
Silent score changes	Scores lock on submission; reopening needs a permission and a reason; versions retained
Silent rule changes	Publication locks rules; changes become approvable diffs with fairness review
Deletion of submitted applications	No deletion path exists; withdrawal and archiving are the available outcomes
Panel members seeing each other's scores	Enforced server-side before submission is complete
Technical administrators making recruitment decisions	System-admin accounts hold three technical permissions and cannot inherit others
Unapproved ERP transfer	Recording a personnel number requires a prior signed approval
Unauthorised export	Report access is permission-scoped; every export is audited
A vacancy advertised without funding	Publication gate
An offer above the approved ceiling	Flagged and returned to the Budget Holder

10.4 Versioning and reconstruction

Beyond the audit log, the platform retains every vacancy version, rule version, score version, approval, override, status change, document version, communication, export and transfer attempt, plus every role and permission change.

Optimistic concurrency guards records that two people may work simultaneously: each carries a lock version, and a stale write is refused with a message telling the user to reload rather than silently overwriting a colleague's decision.

Together these make it possible to reconstruct the complete recruitment process from beginning to end.

10.5 Approvals and conditions

An approval carries a resource type and id, a stage, the assigned approver, the requester, the decision, a comment and a timestamp. Decisions are approve, approve with conditions, return for clarification, or reject.

Approving with conditions creates trackable condition records, each with a description, an owner, a due date, a status, and space for evidence — so a conditional approval is a commitment with follow-through rather than a note in a comment box.

10.6 Work items and service standards

Configurable service standards define expected timelines for staffing request review, budget confirmation, vacancy approval, advertising, longlisting, shortlisting, assessment, interview, reference checking, offer approval, pre-employment clearance, ERP transfer and closure.

From those the platform derives due dates, reminders, escalations, overdue indicators, service-standard reports and delay reasons. Work items surface on each user's queue, scoped to what they can actually action.

11. Security, privacy and data

11.1 Authentication

Password authentication with bcrypt at cost factor 12, chosen as a meaningful brake on offline cracking while staying inside a normal request budget. Passwords are validated for length and composition and capped at 72 UTF-8 bytes, the bcrypt limit.

Additional controls:

- **TOTP multi-factor authentication** (RFC 6238) with recovery codes
- **Automatic lockout** after repeated failures, expiring on its own so a locked-out user recovers without an administrator; a manual administrative lock remains indefinite and separate
- **Per-device sessions** with individual revocation
- **Session versioning**, so a password change or forced sign-out invalidates every existing token
- **Optional SSO** through OIDC

Sessions are JWT cookies. Verification status is deliberately **not** carried in the token — it would go stale the moment someone verified — so a token-only read reports "not verified" and fails closed, and anything depending on it re-reads from the database.

11.2 Authorisation

Three layers, checked in order:

1. **Role** — coarse routing, deciding which workspace a user lands in
2. **Permission** — granular capability, checked per action
3. **Scope** — resource-level assignment, so a hiring manager sees only their own vacancies

Scoped role assignments never become global merely because a caller omitted resource context; the check fails closed until the exact scope is supplied.

System administrators are structurally excluded from operational permissions. If an operational role is mistakenly also assigned to a system-admin account, the platform ignores it.

11.3 Rate limiting and abuse

Distributed rate limiting protects authentication and the public vacancy feed. Public search terms are capped in length. Idempotency keys are required for operations that must not double-execute — recording an ERP transfer, submitting an application — so a retried request returns the original result rather than acting twice.

11.4 Files

Uploaded files are validated for type, size and signature, scanned for viruses, encrypted at rest, and classified by sensitivity as standard, confidential or restricted. Access is checked against ownership and permission on every read. Restricted files are excluded from bulk exports.

11.5 Secrets

Secrets held in the database — TOTP seeds, OAuth access and refresh tokens, PKCE verifiers — are sealed with AES-256-GCM under a key derived from `STORAGE_ENCRYPTION_KEY`. A dump of the database yields no usable credential.

An unopenable token is treated as a changed key: the identity is marked expired with an explanatory message, rather than producing a stream of unexplained authorisation failures.

11.6 Data protection

Requirement	Implementation
Privacy notice	Published, versioned, and the accepted version recorded per candidate
Consent records	Per consent type with the notice version, decision, date and any withdrawal
Data minimisation	Only necessary fields transmitted to any third party
Role-based access	Every read is permission-checked
Encryption	In transit and at rest
Secure upload	Validated, scanned, access-controlled
Retention schedule	Configurable, with recorded retention runs
Correction	Candidates may correct their own profile
Deletion	Requested, reviewed against holds and statutory windows, decided with a recorded basis
Secure archive	Records move to archived state rather than being deleted
Anonymisation	Automatic where configured
Legal hold	Suspends retention and deletion for named records
Breach logging	Operational events with severity
Restricted export	Permission-scoped and audited

11.7 The data model

136 models, grouped by area:

Area	Models	Examples
Identity and permissions	10	User, Role, Permission, RolePermission, UserRole, UserSession, UserMfaSecret, ExternalIdentity
Candidate profile	9	CandidateProfile, CandidateEducation, CandidateEmployment, CandidateLicence, ConsentRecord
Files and documents	2	FileAsset, CandidateDocument
Vacancies and configuration	4	Vacancy, Department, Project, DutyStation

Area	Models	Examples
Staffing requests and funding	6	StaffingRequest, FundingConfirmation, OfferFinancialApproval
Applications	6	Application, ApplicationAnswer, ApplicationProfileSnapshot, ApplicationStageHistory
Scorecards and screening	5	ScorecardTemplate, ScorecardCriterion, CandidateScorecard, ConflictDeclaration
Assessments	4	Assessment, AssessmentQuestion, CandidateAssessment
Interviews	5	Interview, InterviewPanelMember, InterviewScore, InterviewPanelSubmission
References	3	Referee, ReferenceRequest, ReferenceResponse
Selection and approvals	3	SelectionDecision, Approval, ApprovalCondition
Offers	2	Offer, OfferTemplate
Preboarding	22	Packages, forms, documents, policies, courses, tasks, meetings
Readiness and ERP transfer	4	ReadinessCheck, ResumptionRecord, ERPTTransferRecord
Due diligence	1	BackgroundCheck
CV parsing	1	CvParseResult
Electronic signatures	1	ElectronicSignature
Messaging, notifications, audit	7	MessageThread, Notification, AuditLog, AuditChainHead
Recruitment operating system	20	WorkflowDefinition, SlaPolicy, WorkItem, TalentPool, EligibilityRule, LonglistRun
Reliability and governance	18	OutboxMessage, LegalHold, RetentionRun, ComplaintCase, EntityVersion, JobRun

11.8 Talent pools

Approved pools for emergency response, consultants, enumerators, drivers, community mobilisers, health, nutrition, WASH, protection, MEAL, finance and operations, interns and volunteers, and reserve candidates.

Each roster record shows the technical category, preferred location, availability, expected rate or grade, previous assessment results, reference status, background-check status, roster expiry, last verification date and deployment history.

Candidates must consent before being added, and that consent carries an expiry.

12. Emergency recruitment

Emergency recruitment is faster. It is not less controlled. The distinction runs through every part of the design: the platform shortens **timelines** and reuses **pre-approved material**, and it never removes an approval, a check or an audit record.

12.1 Classification

A vacancy is marked as emergency recruitment with a written justification. The classification is then approved separately from the vacancy content, by an HR Manager who does not own the vacancy. An unapproved classification blocks publication.

An emergency staffing request also escalates automatically to an executive approver.

12.2 Shortened service standards

Stage	Emergency	Standard
Staffing request review	4 hours	72 hours
Budget confirmation	4 hours	72 hours
Vacancy approval	4 hours	72 hours
Advertising period	48 hours	336 hours
Longlisting	4 hours	72 hours
Shortlisting	8 hours	120 hours
Assessment	24 hours	168 hours
Interview	24 hours	168 hours
Reference checking	24 hours	120 hours
Offer approval	4 hours	72 hours
Pre-employment clearance	48 hours	240 hours

These drive due dates and escalation. They do not permit a stage to be skipped — only chased sooner.

12.3 The advertising floor

An emergency vacancy must still be open for at least **24 hours**. A vacancy nobody could realistically see was never advertised, and shortening the window to nothing would make the whole exercise a direct appointment wearing a competitive label.

12.4 Controls that are never waived

Five controls hold regardless of urgency, and they are evaluated from evidence already in the record — an emergency hire cannot reach offer approval simply by moving faster than the checks.

- Identity verification
- Safeguarding check
- At least one satisfactory reference
- Budget Holder funding confirmation
- HR Manager offer approval

An explicitly waived check counts as settled, because a waiver is itself a recorded HR Manager decision. A check merely *in progress* does not.

12.5 Accelerations available

- Pre-approved job descriptions attached to the staffing request
- Pre-approved assessment templates
- Candidates drawn from an existing roster, already assessed and reference-checked
- Rapid availability checks against roster records
- Bulk candidate communication
- Same-day or next-day assessment scheduling
- Remote interviews
- Expedited offer preparation

12.6 Post-recruitment compliance review

Every emergency exercise is reviewed afterwards. The review is assembled from what actually happened rather than from a form filled in later, so it cannot be written to look tidier than the record.

It reports the justification, who approved the classification and when, the time from advertising to offer, every acceleration that was used, every non-waivable control that was satisfied, and any findings.

Findings are raised where:

- No written justification was recorded
- The classification was never approved by an authorised officer
- The advertising period fell below the 24-hour minimum
- Any non-waivable control was not completed
- **The exercise was classified as an emergency but used no accelerated provisions at all**

That last one is the most interesting finding the review produces. An emergency classification that changed nothing about how the recruitment ran suggests the classification was unnecessary — and repeated instances suggest it is being used to bypass scrutiny rather than to move quickly.

12.7 Bulk recruitment

For high-volume exercises the platform supports bulk application review, bulk status changes with a preview-then-apply flow and an undo window, bulk communication that never exposes one candidate's address to another, bulk interview scheduling, multiple panels, assessment-centre management, candidate grouping, attendance lists, batch score upload, batch document and reference requests, and batch ERP transfer after final approval.

Every bulk action previews what will change, reports what is ineligible and why, and leaves ineligible records untouched.

12.8 Offline and low-connectivity operation

For field locations with weak internet: lightweight applicant pages, a low-data mode, save-and-continue-later, offline assessment packs, downloadable candidate lists, offline score collection with controlled upload, mobile-friendly forms, and resumption after a connection failure.

13. Dashboards, reporting and communication

13.1 Dashboards

Each role lands on a workspace scoped to what it can actually action.

HR and recruitment officers see vacancies awaiting approval, open vacancies, applications received, the pipeline by stage, longlisting progress, the exception queue, shortlisting progress, upcoming assessments and interviews, pending panel scores, pending references, pending background checks, offers awaiting approval, candidates awaiting ERP transfer, overdue actions and time-to-fill.

HR Managers additionally see longlisting rule changes awaiting approval, longlists awaiting confirmation, recruitment exceptions, vacancies outside service timelines, final recommendations awaiting approval, ERP transfers awaiting approval, compliance gaps, vacancy ageing and recruitment performance by user.

Budget Holders see staffing requests awaiting funding, positions by budget line, proposed salary ceilings, offer variations awaiting approval, funding end dates approaching, and recruitment linked to their projects — and nothing about candidates.

Hiring departments see their submitted requests, recruitment progress, applications assigned for review, pending shortlisting, upcoming interviews, pending scorecards and final recommendations.

Queue counts are only computed for users who hold the permission to act on them, so a dashboard does not run aggregates it will then hide.

13.2 Reports

Thirty-five reports across five groups. Each exports as CSV, PDF or as part of a complete ZIP pack.

Group	Reports
Plan and fund	Staffing requests; funding confirmations
Recruitment	Vacancy pipeline; candidate stage history; longlisting summary; longlisting exceptions and overrides; shortlisting scores; assessment outcomes; interview activity; candidate ranking; selection decisions; reference checks; background-check status
Offer and start	Offer outcomes; pre-start completion; outstanding pre-start items; course completion; readiness checks; start outcomes; ERP handovers; recruitment closure
Governance	Waiver audit; approval decisions; recruitment compliance; electronic signature register; system audit trail; complaint and appeal register; privacy and deletion requests; configuration change approvals
Operations	Time to fill; source of application and hire; work queue and service history; communication register; message delivery and failures; data-quality exceptions

Three deserve particular attention:

Recruitment compliance is a control-by-control view across every vacancy: is a staffing request linked, is it approved, is funding confirmed, is the vacancy approved, are mandatory rules defined, are rules locked, is the longlist confirmed, are shortlisting criteria set, is safeguarding classified, is a recruitment contact recorded. It

answers an auditor's first question in one export.

Time to fill measures from the staffing request, not from advertising — that is the interval the department actually experiences. It reports both.

Longlisting exceptions and overrides shows, for every exception, the automatic outcome, the human decision, the reason code, the justification, whether evidence was attached and whether approval was required. This is where a pattern of overrides becomes visible.

Access is permission-scoped: financial reports require funding authority, due-diligence reports require check-management authority, and restricted content is excluded from exports the requester is not entitled to.

Reports can be scheduled for recurring delivery by email, with the recurrence and every delivery recorded.

13.3 Communication

Templates exist for application confirmation, incomplete application, test invitation, interview invitation, interview rescheduling, document request, reference request, conditional offer, final offer, regret notification, vacancy cancellation, recruitment delay and talent-pool consent.

All communication is logged. **Bulk communication never exposes one candidate's address to another.**

Delivery uses a transactional outbox: a message is written in the same transaction as the action that triggered it, then dispatched by a worker with retries and a dead-letter queue. A failed send is visible in the delivery report rather than silently lost, and an action never half-commits because email was down.

Notifications are idempotent on a daily key, so a reminder job that runs twice does not send twice.

13.4 Analytics

Detailed analytics cover vacancy volume, application volume, qualified application rate, longlisting rejection reasons, shortlisting conversion, assessment and interview pass rates, offer acceptance, candidate withdrawal, recruitment duration, recruitment cost, source effectiveness, diversity at each stage where lawfully monitored, recruitment team performance, panel participation, delayed stages, failed recruitments, repeatedly difficult-to-fill roles, and ERP transfer success and failure.

Dashboards filter by date, department, project, duty station, job family, grade, contract type, HR user, budget holder and vacancy status.

13.5 Search

Permission-aware global search across candidates, contacts, vacancies, projects, departments, duty stations and ERP personnel numbers. Results are filtered to what the searcher may see — a search never confirms the existence of a record the user could not otherwise reach.

14. Configuration, integrations and operations

14.1 No-code configuration

The HR Manager configures routine settings without software development:

Recruitment stages; vacancy approval routes; candidate statuses; longlisting rule templates; shortlisting templates; interview scorecards; assessment templates; offer approval routes; communication templates; recruitment timelines; mandatory documents; ERP transfer requirements.

Reference data — departments, projects, duty stations, vacancy categories, contract types, document types — is administered through dedicated screens.

Configuration changes that affect governance route through a change-request workflow: proposed, reviewed, approved and released, with the proposal and the decision both recorded. Technical configuration affecting security, infrastructure or integrations remains with the technical administrator.

14.2 Integrations

Integration	Status
Microsoft Outlook and Teams	OAuth with PKCE; calendar events, free/busy, Teams meeting links
Google Calendar and Meet	OAuth with PKCE; calendar events, free/busy, Meet links
Zoom	OAuth with PKCE; meeting creation with waiting room enabled
SMTP	Transactional email through the outbox
S3-compatible storage	Encrypted object storage for all uploads
ClamAV	Virus scanning on upload
OIDC	Optional single sign-on
ERP	Manual by design — a generated PDF handover pack, not an API

Calendar providers self-report as unavailable until credentials are configured. Required environment variables:

```
APP_BASE_URL=
MICROSOFT_CLIENT_ID=      MICROSOFT_CLIENT_SECRET=
GOOGLE_CLIENT_ID=        GOOGLE_CLIENT_SECRET=
ZOOM_CLIENT_ID=           ZOOM_CLIENT_SECRET=
STORAGE_ENCRYPTION_KEY=
INTERNAL_EMAIL_DOMAINS=fradfoundation.org
```

Redirect URI per provider:

```
{APP_BASE_URL}/api/integrations/calendar/callback/{microsoft|google|zoom}
```

14.3 Scheduled work

A scheduler drives recurring jobs: vacancy opening and closing, offer expiry, assessment expiry, interview reminders, reference reminders, scheduled report delivery, outbox retries, retention runs, escalations for overdue work items, and daily notification digests.

Each run is recorded with its outcome. Long-running jobs take a lease so two workers cannot process the same batch, and a failed run raises an operational event.

14.4 Technology

Layer	Technology
Framework	Next.js 16, React 18, TypeScript
Database	PostgreSQL via Prisma
Styling	Tailwind CSS
Authentication	JWT sessions with jose, bcrypt hashing, TOTP MFA
Storage	S3-compatible
Email	Nodemailer over SMTP
Testing	Vitest for units, Playwright for end-to-end

14.5 Deployment

```
npx prisma generate
npx prisma migrate deploy
npm run build
npm start
```

The demo seed is disabled when `NODE_ENV=production`; production uses a controlled bootstrap that creates a single administrator with an explicitly supplied password.

Deployment tasks that sit outside the application: provisioning PostgreSQL, configuring SMTP and verifying sender reputation, registering the SSO provider, configuring object storage and virus scanning, arranging worker execution and monitoring, backups, accessibility testing with assistive-technology users, load testing, security and privacy approvals, and disaster-recovery exercises.

14.6 Quality gates

Gate	Command
Type checking	<code>npx tsc --noEmit</code>
Linting	<code>npm run lint</code>
Unit tests	<code>npm test</code>
Integration tests	<code>npm run test:integration</code>
End-to-end tests	<code>npm run test:e2e</code>
Schema integrity	<code>npm run check:schema</code>
Backend-to-frontend coverage	<code>npm run audit:frontend-coverage</code>
Production build	<code>npm run build</code>

`check:schema` runs two static analysers written for this codebase. The first verifies that every Prisma relation is declared on both sides and that no relation points at a model that does not exist. The second parses every `select` and `include` in the application and confirms each field actually exists on the model being queried. Both run without a database or a generated client, which makes them usable in environments where the Prisma engine cannot be downloaded — and both have caught real defects that type checking alone would have missed until runtime.

15. Component index

Generated from the codebase. Every screen, endpoint and module the platform contains.

Component type	Count
Screens (pages)	103
API endpoints	132
Domain modules	75
UI components	65
Database models	136

15.1 Screens

Careers portal (2)

- `/careers`
- `/careers/[reference]`

Public (11)

- `/complaints`
- `/forgot-password`
- `/guidance`
- `/page.tsx`
- `/privacy`
- `/public/reference/[token]`
- `/recruitment-faq`
- `/report-fraud`
- `/reset-password`
- `/terms`
- `/verify-email`

Authentication (2)

- `/auth/login`
- `/auth/register`

Candidate portal (27)

- `/candidate/accommodations`
- `/candidate/applications`
- `/candidate/applications/[id]`
- `/candidate/applications/[id]/receipt`
- `/candidate/applications/apply`
- `/candidate/assessments/[id]`
- `/candidate/complaints`
- `/candidate/dashboard`
- `/candidate/interviews`
- `/candidate/messages`
- `/candidate/offers/[id]`
- `/candidate/preboarding`
- `/candidate/preboarding/courses`
- `/candidate/preboarding/documents`
- `/candidate/preboarding/forms`
- `/candidate/preboarding/meetings`
- `/candidate/preboarding/policies`
- `/candidate/preboarding/reporting-information`
- `/candidate/preboarding/tasks`
- `/candidate/profile`
- `/candidate/profile/documents`
- `/candidate/profile/education`
- `/candidate/profile/employment`
- `/candidate/profile/licences`
- `/candidate/profile/personal`
- `/candidate/settings`
- `/candidate/tasks`

Recruitment workspace (34)

- `/recruitment/accommodations`
- `/recruitment/applications`
- `/recruitment/applications/[id]`
- `/recruitment/applications/[id]/handover`
- `/recruitment/approvals`
- `/recruitment/assessments`
- `/recruitment/audit`
- `/recruitment/background-checks`

- `/recruitment/communications`
- `/recruitment/complaints`
- `/recruitment/dashboard`
- `/recruitment/erp-transfers`
- `/recruitment/funding`
- `/recruitment/insights`
- `/recruitment/interviews`
- `/recruitment/longlisting`
- `/recruitment/longlisting/exceptions`
- `/recruitment/offers`
- `/recruitment/operations`
- `/recruitment/preboarding`
- `/recruitment/preboarding/[id]`
- `/recruitment/quality`
- `/recruitment/references`
- `/recruitment/reports`
- `/recruitment/search`
- `/recruitment/selections`
- `/recruitment/settings`
- `/recruitment/staffing-requests`
- `/recruitment/talent-pools`
- `/recruitment/vacancies`
- `/recruitment/vacancies/[id]`
- `/recruitment/vacancies/[id]/edit`
- `/recruitment/vacancies/new`
- `/recruitment/work`

Administration (27)

- `/admin/assessment-bank`
- `/admin/automations`
- `/admin/configuration-releases`
- `/admin/contract-types`
- `/admin/courses`
- `/admin/deletion-requests`
- `/admin/departments`
- `/admin/document-requirements`
- `/admin/document-types`
- `/admin/duty-stations`

- `/admin/forms`
- `/admin/fraud-reports`
- `/admin/governance`
- `/admin/interview-questions`
- `/admin/notification-templates`
- `/admin/operating-model`
- `/admin/permissions`
- `/admin/policies`
- `/admin/preboarding-packages`
- `/admin/projects`
- `/admin/roles`
- `/admin/scorecards`
- `/admin/system-settings`
- `/admin/tasks`
- `/admin/templates`
- `/admin/users`
- `/admin/vacancy-categories`

15.2 API endpoints

Administration (9)

- `/api/admin/automations`
- `/api/admin/configuration-builder`
- `/api/admin/configuration-releases`
- `/api/admin/deletion-requests`
- `/api/admin/fraud-reports`
- `/api/admin/generic`
- `/api/admin/governance`
- `/api/admin/operating-model`
- `/api/admin/users/[id]`

Authentication (12)

- `/api/auth/forgot-password`
- `/api/auth/login`
- `/api/auth/logout`
- `/api/auth/mfa`
- `/api/auth/mfa/challenge`
- `/api/auth/register`
- `/api/auth/reset-password`
- `/api/auth/session`
- `/api/auth/sessions`
- `/api/auth/sso/callback`
- `/api/auth/sso/start`
- `/api/auth/verify-email`

Calendar (1)

- `/api/calendar/interviews/[id]`

Candidate (28)

- `/api/candidate/accommodations`
- `/api/candidate/account`
- `/api/candidate/applications`
- `/api/candidate/applications/[id]`
- `/api/candidate/assessments/[id]`
- `/api/candidate/assessments/[id]/answers`
- `/api/candidate/assessments/[id]/answers/review`

- `/api/candidate/assessments/[id]/start`
- `/api/candidate/assessments/[id]/submit`
- `/api/candidate/documents`
- `/api/candidate/documents/[id]`
- `/api/candidate/education`
- `/api/candidate/education/[id]`
- `/api/candidate/employment`
- `/api/candidate/employment/[id]`
- `/api/candidate/interviews/[id]/respond`
- `/api/candidate/licences`
- `/api/candidate/licences/[id]`
- `/api/candidate/offers/[id]`
- `/api/candidate/offers/[id]/respond`
- `/api/candidate/preboarding`
- `/api/candidate/preboarding/actions`
- `/api/candidate/preboarding/confirm-start-date`
- `/api/candidate/preboarding/courses/[id]/certificate`
- `/api/candidate/privacy/export`
- `/api/candidate/profile`
- `/api/candidate/profile-items`
- `/api/candidate/saved-searches`

Cases (2)

- `/api/complaints`
- `/api/complaints/[id]/comments`

Files (2)

- `/api/assets/download/[id]`
- `/api/assets/upload`

Integrations (2)

- `/api/integrations/calendar/callback/[provider]`
- `/api/integrations/calendar/connect`

Messaging (1)

- `/api/messages`

Operations (1)

- `/api/health`

Public (5)

- `/api/public/fraud-reports`
- `/api/public/reference/resolve`
- `/api/public/reference/submit`
- `/api/public/vacancies`
- `/api/public/vacancies/[reference]`

Recruitment (68)

- `/api/recruitment/accommodations`
- `/api/recruitment/applications`
- `/api/recruitment/applications/[id]`
- `/api/recruitment/applications/[id]/assign-reviewer`
- `/api/recruitment/applications/[id]/conflict`
- `/api/recruitment/applications/[id]/documentation`
- `/api/recruitment/applications/[id]/erp-transfer`
- `/api/recruitment/applications/[id]/erp-transfer/approve`
- `/api/recruitment/applications/[id]/erp-transfer/pack`
- `/api/recruitment/applications/[id]/handover-summary`
- `/api/recruitment/applications/[id]/notes`
- `/api/recruitment/applications/[id]/referees`
- `/api/recruitment/applications/[id]/resumption`
- `/api/recruitment/applications/[id]/stage`
- `/api/recruitment/applications/assisted`
- `/api/recruitment/applications/bulk-actions`
- `/api/recruitment/applications/bulk-actions/[id]/undo`
- `/api/recruitment/applications/bulk-export`
- `/api/recruitment/applications/bulk-stage-change`
- `/api/recruitment/approvals`
- `/api/recruitment/assessments`
- `/api/recruitment/assessments/[id]`
- `/api/recruitment/assessments/[id]/invite`
- `/api/recruitment/background-checks`
- `/api/recruitment/candidate-assessments/[id]/answers`
- `/api/recruitment/candidate-assessments/[id]/mark`
- `/api/recruitment/candidate-assessments/[id]/reset`

- /api/recruitment/complaints
- /api/recruitment/data-quality/merges
- /api/recruitment/eligibility
- /api/recruitment/interviews
- /api/recruitment/interviews/[id]
- /api/recruitment/interviews/[id]/confirm-panel
- /api/recruitment/interviews/[id]/invite
- /api/recruitment/interviews/[id]/panel/[memberId]/reopen
- /api/recruitment/interviews/[id]/panel/[memberId]/resolve-conflict
- /api/recruitment/interviews/[id]/scores
- /api/recruitment/interviews/availability
- /api/recruitment/longlisting/exceptions
- /api/recruitment/longlisting/rules
- /api/recruitment/longlisting/runs
- /api/recruitment/offers
- /api/recruitment/offers/[id]
- /api/recruitment/offers/[id]/actions
- /api/recruitment/offers/[id]/preview
- /api/recruitment/preboarding/[id]
- /api/recruitment/preboarding/[id]/manage
- /api/recruitment/preboarding/clearance
- /api/recruitment/referees/[id]/send-reminder
- /api/recruitment/referees/[id]/send-request
- /api/recruitment/reference-responses/[id]/verify
- /api/recruitment/reports/export
- /api/recruitment/reports/schedules
- /api/recruitment/scorecards
- /api/recruitment/scorecards/[id]
- /api/recruitment/scorecards/[id]/reopen
- /api/recruitment/search
- /api/recruitment/selections
- /api/recruitment/staffing-requests
- /api/recruitment/staffing-requests/[id]/actions
- /api/recruitment/staffing-requests/[id]/funding
- /api/recruitment/talent-pools
- /api/recruitment/vacancies
- /api/recruitment/vacancies/[id]
- /api/recruitment/vacancies/[id]/actions

- `/api/recruitment/vacancies/[id]/documentation`
- `/api/recruitment/vacancies/[id]/emergency-review`
- `/api/recruitment/work-items/[id]`

Scheduled jobs (1)

- `/api/cron/process-schedules`

15.3 Domain modules

Module	Purpose
<code>anonymisation.ts</code>	Anonymised longlisting and shortlisting (End_to_End.md §28.3)
<code>application-reference.ts</code>	—
<code>application-stages.ts</code>	Every internal application stage, in pipeline order
<code>approvals.ts</code>	Choose an approver who is independent of the requester
<code>audit.ts</code>	Verify the tamper-evident audit chain
<code>auth.ts</code>	Resolve the JWT signing secret. No insecure fallback: the process must be
<code>authz.ts</code>	Well-known Prisma error codes mapped to the response a caller can act on
<code>automations.ts</code>	—
<code>background-checks.ts</code>	Background and due-diligence checks (End_to_End.md §16, §28.11)
<code>background-jobs.ts</code>	—
<code>calendar-identity.ts</code>	Storage and lifecycle for a user's calendar OAuth grant (§28.15)
<code>calendar-providers.ts</code>	Calendar and video-meeting providers (End_to_End.md §28.15)
<code>candidate-preboarding.ts</code>	—
<code>candidate-status.ts</code>	—
<code>candidate-tasks.ts</code>	—
<code>complaint-workflow.ts</code>	—
<code>concurrency.ts</code>	—
<code>configuration-releases.ts</code>	—
<code>cv-parser.ts</code>	Structured CV parsing (End_to_End.md §28.1)
<code>deterministic-shuffle.ts</code>	—
<code>eligibility.ts</code>	Longlisting execution (End_to_End.md §11.3)
<code>emergency-recruitment.ts</code>	Emergency recruitment (End_to_End.md §28.7)
<code>erp-handover.ts</code>	ERP handover (End_to_End.md §19)
<code>errors.ts</code>	—
<code>export-files.ts</code>	—
<code>form-template-fields.ts</code>	—
<code>form-template.ts</code>	—
<code>home-route.ts</code>	One source of truth for the first page a signed-in user sees
<code>idempotency.ts</code>	—
<code>internal-identity.ts</code>	Internal candidate identity (End_to_End.md §28.8)

Module	Purpose
<code>job-alerts.ts</code>	Emails candidates when a vacancy matching one of their saved searches opens
<code>logout.ts</code>	Automatic account lockout after consecutive failed sign-in attempts
<code>logger.ts</code>	Tiny structured logger. Emits single-line JSON so logs are greppable and can
<code>longlisting-rules.ts</code>	Rule-based longlisting (End_to_End.md §11)
<code>mailer.ts</code>	Email transport. Uses nodemailer over SMTP when SMTP_* env vars are set and
<code>message-template-fields.ts</code>	—
<code>message-template.ts</code>	—
<code>notifications.ts</code>	—
<code>offer-document.ts</code>	—
<code>offer-template-fields.ts</code>	—
<code>offer-template.ts</code>	—
<code>oidc.ts</code>	—
<code>outbox.ts</code>	—
<code>pagination.ts</code>	One pagination contract for every list endpoint
<code>policy-template.ts</code>	—
<code>preboarding.ts</code>	—
<code>prisma.ts</code>	—
<code>profile-completion.server.ts</code>	—
<code>profile-completion.ts</code>	—
<code>qr.ts</code>	Minimal QR code encoder producing an inline SVG
<code>rate-limit.ts</code>	Minimal in-memory sliding-window rate limiter. Suitable for a single-instance
<code>rbac.ts</code>	System administration is a technical control-plane role. It must never
<code>recruitment-access.ts</code>	Resolve access against the concrete application. Permission checks alone are
<code>recruitment-role-policy.ts</code>	§3.7 Funding authority sits with the Budget Holder — the person with authority
<code>recruitment-scoring.server.ts</code>	—
<code>recruitment-scoring.ts</code>	Pure selection-scoring maths. This module must not import Prisma: it is
<code>references.ts</code>	—
<code>retention.ts</code>	Apply the approved short-lived-record policy as one serializable database
<code>roles.ts</code>	Role classification shared by Edge middleware, server routes and client UI
<code>s3.ts</code>	File storage adapter. Persists bytes to a local storage root

Module	Purpose
<code>scheduled-report.ts</code>	—
<code>search.ts</code>	PostgreSQL full-text search over vacancies and candidates
<code>secret-box.ts</code>	AES-256-GCM envelope for short secrets held in the database (currently TOTP)
<code>session.ts</code>	Single place where a signed-in session is minted
<code>signature-policy.ts</code>	Signature policy and hashing (End_to_End.md §28.10)
<code>signatures.ts</code>	Electronic approvals and signatures (End_to_End.md §28.10)
<code>simple-pdf.ts</code>	A generic branded, sectioned A4 document
<code>staffing-request.ts</code>	Staffing request lifecycle (End_to_End.md §5)
<code>state-machine.ts</code>	Application & vacancy state transition rules (README §42)
<code>tokens.ts</code>	—
<code>totp.ts</code>	TOTP (RFC 6238) over HMAC-SHA1 with 6 digits and a 30-second step — the
<code>utils.ts</code>	The timezone every rendered date is expressed in
<code>validation.ts</code>	Parse and validate a request body against a Zod schema
<code>virus-scan.ts</code>	File anti-virus scan hook
<code>work-items.ts</code>	Materialises actionable work from authoritative workflow records. Upserts

15.4 Database models

- 1. Identity & Permissions** — [User](#), [UserMfaSecret](#), [UserRecoveryCode](#), [UserSession](#), [SavedSearch](#), [ExternalIdentity](#), [Role](#), [Permission](#), [RolePermission](#), [UserRole](#)
- 2. Candidate Profile** — [CandidateProfile](#), [ConsentRecord](#), [DataDeletionRequest](#), [CandidateEducation](#), [CandidateEmployment](#), [CandidateLicence](#), [CandidateCertification](#), [CandidateSkill](#), [CandidateLanguage](#)
- 3. File Storage & Documents** — [FileAsset](#), [CandidateDocument](#)
- 4. Vacancies & Config** — [Department](#), [VacancyCategory](#), [Project](#), [DutyStation](#)
- 4a. Staffing Requests & Funding (End_to_End.md §5, §3.7)** — [StaffingRequest](#), [FundingConfirmation](#), [OfferFinancialApproval](#), [Vacancy](#), [VacancyQuestion](#), [VacancyRequiredDocument](#)
- 5. Applications** — [Application](#), [ApplicationProfileSnapshot](#), [ApplicationAnswer](#), [ApplicationFile](#), [ApplicationNote](#), [ApplicationStageHistory](#)
- 6. Scorecards & Screening** — [ScorecardTemplate](#), [ScorecardCriterion](#), [CandidateScorecard](#), [CandidateCriterionScore](#), [ConflictDeclaration](#)
- 7. Assessments** — [Assessment](#), [AssessmentQuestion](#), [CandidateAssessment](#), [CandidateAssessmentAnswer](#)
- 8. Interviews** — [Interview](#), [InterviewPanelMember](#), [InterviewQuestion](#), [InterviewScore](#), [InterviewPanelSubmission](#)
- 9. References** — [Referee](#), [ReferenceRequest](#), [ReferenceResponse](#)
- 10. Selection & Approvals** — [SelectionDecision](#), [Approval](#), [ApprovalCondition](#)
- 11. Offers** — [OfferTemplate](#), [Offer](#)
- 12. Preboarding Packages & Requirements** — [PreboardingPackage](#), [CandidatePreboarding](#), [CandidatePreboardingPackage](#)
- 13. Preboarding Forms** — [PreboardingFormTemplate](#), [PackageForm](#), [CandidatePreboardingForm](#)
- 14. Document Requirements** — [DocumentRequirement](#), [PackageDocumentRequirement](#), [CandidateRequiredDocument](#), [CandidateRequiredDocumentVersion](#)
- 15. Policies & Signatures** — [PolicyDocument](#), [PackagePolicy](#), [CandidatePolicyAcknowledgement](#)
- 16. Compulsory Courses** — [Course](#), [CourseContent](#), [CourseQuizQuestion](#), [PackageCourse](#), [CandidateCourse](#), [CandidateCourseContentProgress](#), [CandidateCourseAttempt](#)
- 17. Pre-resumption Tasks & Info** — [PreboardingTaskTemplate](#), [PackageTask](#), [CandidatePreboardingTask](#), [CandidateInformationItem](#)
- 18. Meetings & Orientation** — [PreboardingMeeting](#)
- 19. Readiness, Resumption & ERP Transfer** — [ReadinessCheck](#), [ReadinessConfirmation](#), [ResumptionRecord](#), [ERPTransferRecord](#)
- 19a. Background & Due-Diligence Checks (§16, §28.11)** — [BackgroundCheck](#)
- 19b. CV Parsing (§28.1)** — [CVParseResult](#)
- 19c. Electronic Signatures (§28.10)** — [ElectronicSignature](#)

20. Messaging, Notifications & Audit — `MessageThread`, `Message`, `Notification`, `AuditLog`, `AuditChainHead`, `FraudReport`, `SystemSetting`

22. Recruitment operating system — `WorkflowDefinition`, `WorkflowVersion`, `WorkflowTransitionRule`, `SlaPolicy`, `WorkItem`, `TalentPool`, `TalentPoolMember`, `AccommodationRequest`, `IntegrationConnection`, `IntegrationIdentity`, `IntegrationOAuthState`, `AvailabilityWindow`, `ConfigurationChangeRequest`, `AutomationControl`, `AutomationActionLog`, `BulkActionRun`, `CandidateMergeReview`, `NotificationTemplate`, `ContractType`, `DocumentType`

21. Reliability, Governance & Case Management — `OutboxMessage`, `IdempotencyRecord`, `LegalHold`, `RetentionRun`, `AccessReview`, `ComplaintCase`, `ComplaintComment`, `ComplaintAttachment`, `EntityVersion`, `EligibilityRule`, `EligibilityRuleChange`, `LonglistRun`, `EligibilityEvaluation`, `OperationalEvent`, `JobRun`, `JobLease`, `RateLimitBucket`, `ScheduledReport`